

NerveML

Neural Data Privacy & Validation Scanner

Prototype Product Definition and Engineering Specification

Core product promise

Before neural data is shared, sold, or used to train AI, NerveML tests what sensitive information can be inferred from it and whether apparent model performance survives leakage-safe validation.

Document purpose	Primary audience	Prototype status
Translate the company thesis into an engineer-ready MVP specification	Founders, ML engineers, neuroscientists, design partners, YC reviewers	Buildable proof-of-concept; not a clinical, legal, or regulatory certification system

Prepared for rapid prototype implementation

1. Executive Summary

NerveML is a validation layer for AI trained on neural data. The first prototype focuses on EEG emotion datasets and demonstrates two risks that neural-data teams routinely struggle to measure:

- Unintended sensitive inference: whether a neural dataset contains information from which a model can infer a sensitive attribute such as self-reported emotional valence.
- Invalid or inflated evaluation: whether apparent decoding performance is caused by subject leakage, repeated measurements, session structure, artifacts, or other confounds rather than a generalizable neural representation.

The MVP does not attempt to “read true emotions,” certify compliance, or solve all neural privacy. Its purpose is to produce an immediately understandable, technically honest demonstration: the same EEG dataset can appear highly predictive under a naive trial-level split and become much weaker when entire subjects are held out. NerveML converts that discrepancy into a structured validation and privacy-risk report.

One-sentence YC version

We built a working prototype that detects subject leakage and unintended emotion inference in EEG datasets.

2. Background and Origin of the Problem

2.1 The market shift

Neural and human-sensing datasets are beginning to move beyond isolated academic experiments into wearable devices, sleep technology, digital biomarkers, clinical neurotechnology, robotics data pipelines, and emerging brain-computer interface systems. These teams increasingly collect repeated, high-dimensional signals from the same people and use machine learning to extract useful representations.

However, the same signals can encode more than the intended task. A dataset collected for motor control, sleep staging, device calibration, or human-robot training may also contain information related to identity, emotional state, cognitive workload, disease, medication, fatigue, or recording conditions.

2.2 Why ordinary de-identification is insufficient

Removing names, emails, and patient IDs does not necessarily remove inferential privacy risk. A model may still identify a person or infer a sensitive state from the signal itself. Neural privacy is therefore not only a question of direct identifiers; it is a question of what attributes remain decodable from the underlying data.

2.3 Why ordinary ML validation is insufficient

Neural data contains repeated trials, sessions, stimuli, devices, and measurements nested within people. A random train/test split can place trials from the same person in both sets. The model can then exploit stable person-specific patterns and appear to generalize even though it fails on a new person. Similar leakage can occur across sessions, sites, devices, and repeated stimuli.

Founder insight

The central problem is not merely whether a model predicts correctly. It is whether the model learned the scientific construct and intended signal that the team believes it learned.

3. Product Definition and Positioning

3.1 Category

Recommended category: Neural data validation infrastructure.

Supporting descriptions:

- Sensitive inference scanner for neural data
- Privacy and validity layer for neural AI
- Evaluation infrastructure for AI trained on neural and human-sensing data

3.2 What the product is

NerveML ingests a neural dataset and associated metadata, runs a standardized suite of leakage-safe model evaluations and sensitive-attribute probes, and generates a report explaining what can be inferred, how robust the result is, which failure modes were detected, and what remediation is recommended.

3.3 What the product is not

- Not a consumer emotion-reading product.
- Not a claim that EEG reveals a person's complete or "true" emotional state.
- Not a replacement for IRB review, legal counsel, medical-device validation, or regulatory certification.
- Not initially a large-scale neural foundation model or distributed GPU infrastructure company.
- Not a one-off scientific paper; the goal is a repeatable software workflow.

3.4 Wedge, platform, and long-term vision

Layer	Scope	Purpose
Initial wedge	EEG emotion/valence datasets	Show unintended sensitive inference and subject-leakage detection with a concrete, understandable example.
Product platform	Neural model and dataset validation	Expand to identity, sleep, diagnosis, medication, motor state, device, session, and site leakage.
Long-term vision	Validation infrastructure for high-dimensional human data	Become the trust layer for neural, physiological, behavioral, and scientific AI systems.

4. Target Users and Jobs to Be Done

User segment	Current workflow/problem	Job NerveML performs	Early willingness to pay
Neurotech and EEG startups	Custom notebooks; uncertain cross-subject performance; pressure from customers and investors	Validate datasets and models before external claims, pilots, or data sharing	Medium
Sleep and wearable teams	Signals mix sleep stage, movement, identity, and affective state	Test whether models rely on intended physiology or nuisance variables	Medium

Human-data / robotics providers	Need to package and sell human data without exposing unintended personal attributes	Scan datasets before downstream delivery	Medium to high if tied to enterprise procurement
Hospitals and clinical researchers	Patient-level leakage, cross-site failures, secondary-use concerns	Generate patient-aware validation and risk documentation	High, but slower sales
Pharma and digital biomarker teams	Biomarkers may fail across cohorts or depend on site/device confounds	Validate generalization before costly trials	High
Academic laboratories	Reviewers question leakage, confounds, and generalizability	Standardize analyses and generate reproducible reports	Low; useful for adoption and design partnerships

4.1 The true needs the prototype is testing

- “I do not know whether my model learned emotion, subject identity, movement, or recording artifacts.”
- “My result looks strong within the dataset but collapses on a new patient or participant.”
- “Before sharing or selling this dataset, I need to know what unintended attributes it reveals.”
- “I need a defensible report for a reviewer, customer, hospital, privacy team, or investor.”
- “Our team repeatedly rebuilds the same validation checks manually.”

5. Core Product Hypotheses

Hypothesis	How the prototype tests it	Success signal
H1: Neural datasets contain unintended sensitive information	Train a valence-sensitive attribute probe on EEG features	Strict grouped performance is measurably above a label-permutation baseline
H2: Naive splits can inflate performance	Compare trial-random CV with subject-held-out CV	Material performance gap and an interpretable warning
H3: Users value an automated report more than another notebook	Convert analyses into a standardized risk report	Design partners can understand and act on the report without reading code
H4: The workflow generalizes beyond emotion	Design metadata and tests around arbitrary target and grouping columns	Same pipeline can later test identity, sleep, medication, diagnosis, motor, session, and site
H5: Validation can become recurring infrastructure	Structure output as machine-readable results and repeatable scans	Easy integration into future data/model pipelines

6. MVP Scope

6.1 Required MVP capabilities

1. Load one supported EEG emotion dataset with subject, trial, signal/features, and valence labels.
2. Provide a dataset summary and basic integrity checks.
3. Extract interpretable channel-by-frequency-band features or accept a precomputed feature table.
4. Train a baseline sensitive-attribute model for high versus low valence.
5. Compare naive trial-random evaluation with subject-grouped evaluation.
6. Compute the generalization/leakage gap.

7. Run a label-permutation null test.
8. Generate risk flags, restrained interpretation, and remediation recommendations.
9. Export results as JSON/CSV and display them in a simple web interface or report.

6.2 Optional stretch capabilities

- Subject identity inference attack.
- EOG/EMG or movement-only confound baseline.
- Stimulus-held-out, session-held-out, site-held-out, and device-held-out validation.
- Probability calibration and confidence intervals across subjects.
- Automated PDF report generation.
- Upload of a customer-provided trained model in addition to dataset-only analysis.

6.3 Explicitly out of scope for the first prototype

- Deep neural networks, GPU clusters, distributed training, or foundation-model pretraining.
- Differential privacy, federated learning, adversarial unlearning, or production-grade de-identification.
- Clinical diagnosis, treatment recommendations, or legal compliance certification.
- Support for every neural modality and file format.

7. End-to-End User Workflow

Step	User action	System action	Primary output
1. Ingest	Upload a supported dataset or select demo dataset	Validate schema and metadata	Dataset inventory and warnings
2. Configure	Choose signal/feature columns, subject column, and sensitive target	Infer evaluation unit and class distribution	Analysis plan
3. Probe	Run valence-sensitive attribute model	Train baseline classifier within leakage-safe pipelines	Naive and grouped metrics
4. Validate	Run subject-held-out CV and permutation test	Evaluate generalization and null distribution	Leakage gap and confidence
5. Interpret	Review features and warnings	Map results to bounded claims	Risk flags and feature associations
6. Remediate	Select recommended actions	Provide checklist for stronger evidence	Next-step tests
7. Export	Download result	Serialize report and metrics	JSON/CSV and one-page risk card

8. Data Contract and Input Requirements

8.1 Minimum required fields

Field	Description	Example
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subject_id	Independent person or patient identifier	s01
trial_id	Trial or recording unit identifier	trial_004
target_label	Sensitive attribute to test; first MVP uses valence	high_valence
signal or features	EEG epochs or precomputed channel-band features	Fp1_beta = -3.14
sampling metadata	Sampling rate and channel names when raw signals are used	128 Hz; 32 EEG channels

8.2 Recommended metadata for later versions

- session_id, site_id, device_id, stimulus_id, recording date, task condition, EOG/EMG channels, reaction time, medication status, sleep stage, consent and intended-use metadata.

8.3 Label language

The UI and report must use bounded language. For DEAP-like data, the target is “self-reported valence label,” not “true emotion.” The product tests whether a label is statistically inferable under the stated protocol; it does not establish psychological ground truth or causal neural representation.

9. Technical Architecture

Component	Recommended first implementation	Responsibility
Data loader	Python; pandas/numpy; optional MNE	Load demo/raw EEG or feature tables; validate schema
Feature pipeline	MNE Welch PSD or supplied features	Produce log band power for delta/theta/alpha/beta/gamma by channel
Model layer	scikit-learn RandomForest and/or LogisticRegression	Provide fast, interpretable baselines
Validation engine	StratifiedKFold + GroupKFold/LeaveOneGroupOut	Compare naive and independent-unit evaluation
Null testing	Within-subject label permutation	Estimate whether grouped performance exceeds chance
Interpretability	Standardized coefficients and feature importance	Rank channel-band associations with caution
Risk rules	Simple transparent heuristics	Convert metrics into prototype warnings
Interface	Streamlit	Upload/configure/run/results/export
Artifacts	JSON, CSV, PNG, optional PDF	Enable demo, reproducibility, and future API integration

9.1 Recommended repository structure

```
nerveml/  
  app.py                # Streamlit interface  
  loaders.py            # dataset loaders and schema validation  
  features.py           # EEG feature extraction  
  models.py             # baseline model definitions  
  validation.py         # trial and grouped CV, permutation tests  
  risk_rules.py         # warnings and bounded interpretations  
  report.py             # JSON/CSV/PDF report generation  
  configs/  
    deap.yaml  
  tests/  
  sample_data/  
  outputs/
```

10. Analysis Pipeline: Detailed Steps

10.1 Dataset integrity scan

Count subjects and trials, check missing values and duplicates, inspect label balance, verify that subject IDs are available. Stop or warn when an independent grouping variable is missing.

10.2 Feature preparation

For raw EEG, compute log band-power features by channel. Fit any learned preprocessing, imputation, scaling, or feature selection only inside the training folds.

10.3 Sensitive attribute probe

Train a baseline model to predict high versus low self-reported valence. Use balanced accuracy and ROC-AUC rather than raw accuracy alone.

10.4 Naive trial-random evaluation

Run shuffled stratified cross-validation at the trial level. Label this clearly as a weaker evaluation that allows the same subject to appear in train and test.

10.5 Subject-held-out evaluation

Run GroupKFold or leave-one-subject-out evaluation. All trials from a subject remain entirely in train or test.

10.6 Leakage/generalization gap

Calculate naive score minus grouped score. Treat a large positive gap as evidence that within-subject information inflates performance; do not automatically claim malicious leakage.

10.7 Permutation null

Shuffle labels within subject and rerun the complete grouped pipeline. Compare observed grouped score against the null distribution.

10.8 Feature association output

Rank channel-band features from logistic coefficients and/or random forest importance. State that associations are model-dependent and non-causal.

10.9 Risk interpretation

Combine grouped performance, naive-grouped gap, null-test evidence, and data-quality warnings into a small set of transparent flags.

10.10 Remediation

Recommend appropriate next tests: subject/session/site/stimulus-held-out evaluation, artifact-only models, external validation, privacy-removal experiments, and consent review.

11. Metrics and Prototype Risk Rules

11.1 Required metrics

Metric	Why it matters
Balanced accuracy	More robust than raw accuracy when high/low valence classes are imbalanced.
ROC-AUC	Measures ranking discrimination across decision thresholds.
Grouped fold mean and standard deviation	Shows variability when independent subjects are held out.
Leakage/generalization gap	Quantifies inflation between trial-random and subject-held-out evaluation.
Permutation p-value and null distribution	Tests whether grouped performance exceeds a pipeline-specific chance baseline.
Per-subject performance when possible	Prevents a global average from hiding severe failures for some participants.

11.2 Illustrative prototype heuristics

These rules are only for the MVP user experience and must be labeled as prototype heuristics, not legal, clinical, or scientific standards.

Condition	Prototype flag	Interpretation
Grouped AUC < 0.55	Low or inconclusive sensitive inference	No convincing cross-subject evidence under this test.
Grouped AUC 0.55-0.65	Moderate inference signal	Some information may be present; requires stronger validation.
Grouped AUC > 0.65 and permutation evidence	Elevated inference signal	Sensitive label appears inferable across held-out subjects under this protocol.
Naive minus grouped balanced accuracy ≥ 0.10	Possible subject dependence	Performance may rely substantially on person-specific structure.
Naive minus grouped balanced accuracy ≥ 0.20	High generalization warning	Naive result is likely misleading for cross-person claims.
Permutation p ≥ 0.05	Insufficient evidence	Observed grouped result does not clearly

		exceed the null at the selected permutation count.
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12. Output: Neural Dataset Risk Report

The report is the product-facing layer. It should be understandable to an engineer, scientist, privacy lead, or investor without requiring them to inspect the notebook.

Report section	Contents
Dataset summary	Modality, subjects, trials, channels/features, label definition, missingness, class balance
Intended inference	What the dataset/model was designed to predict, when supplied
Sensitive attribute tested	Valence in the first MVP; later identity, sleep, diagnosis, medication, motor state, etc.
Naive result	Trial-random metrics with a clear warning about repeated subjects
Strict result	Subject-held-out metrics and fold variability
Generalization gap	Difference between naive and strict evaluation
Null evidence	Permutation result and limitations
Potential feature associations	Top channel-band features, explicitly non-causal
Risk flags	Plain-language warnings with transparent rules
Permitted claim	Example: "self-reported valence was weakly decodable across held-out subjects in this dataset"
Prohibited/unsupported claim	Example: "the system reads a person's true emotion"
Remediation	Next validation, confound, privacy, and documentation steps

Example headline result

Valence was decodable under random trial validation, but performance declined substantially when entire subjects were held out. The current evidence does not support a general cross-person emotion decoder and suggests elevated subject-dependence risk.

13. Streamlit Demo Specification

Screen	Elements	Demo purpose
1. Dataset	Demo dataset selector or upload; schema summary	Establish that the product scans a real data asset
2. Configuration	Target label; subject/group column;	Make the evaluation assumptions explicit

	model; number of folds/permutations	
3. Running	Progress indicators for feature extraction, naive CV, grouped CV, null test	Show a repeatable automated workflow
4. Results	Four main cards: naive score, grouped score, gap, sensitive-inference risk	Deliver the core insight in seconds
5. Evidence	Fold chart, permutation distribution, top features	Support technical credibility
6. Report	Risk flags, bounded claims, recommendations, download buttons	Turn analysis into an actionable product

13.1 Four primary result cards

- Naive trial-level performance
- Held-out-subject performance
- Generalization/leakage gap
- Sensitive inference risk level

14. Engineering Acceptance Criteria

Priority	Acceptance criterion
P0	The application runs end-to-end on one public or synthetic multi-subject EEG dataset.
P0	It predicts binary valence with a Random Forest or Logistic Regression baseline.
P0	It reports both trial-random and subject-grouped cross-validation.
P0	No learned preprocessing is fit on the test folds.
P0	It computes and displays the performance gap.
P0	It generates at least one automatic risk warning and one bounded interpretation.
P0	It exports a machine-readable audit_report.json.
P1	It runs a label-permutation test and displays the null distribution.
P1	It exports top feature associations and a plot.
P1	It provides a Streamlit results page and downloadable report.
P2	It supports identity inference or artifact-only baselines.

14.1 Definition of done for the YC demo

- A two-minute screen recording can show dataset -> scan -> naive versus grouped difference -> risk report.
- All displayed numbers come from a real executed pipeline, not mock values.

- The product language distinguishes label prediction from true mental-state reading.
- The founder can explain why a normal notebook is insufficient and how this becomes a repeated validation workflow.

15. Suggested Founder and Engineer Responsibilities

Role	Responsibilities
ML / neuroscience engineer	Dataset loading; feature extraction; baseline models; grouped CV; permutation tests; artifact export; tests
Founder / product lead	Product narrative; risk taxonomy; report language; UI flow; customer interview framing; demo script; YC answers
Joint	Choose dataset; verify scientific assumptions; interpret results; prevent overclaiming; identify design partners

16. Two-Minute Demo Script

Time	Narration / action
0:00-0:20	Neural datasets contain overlapping information about emotion, identity, movement, and recording conditions. Standard model evaluation can look accurate while memorizing the people in the dataset.
0:20-0:40	Open NerveML and load a multi-subject EEG dataset. Select self-reported valence as the sensitive attribute and subject_id as the independent grouping unit.
0:40-1:10	Run the scan. Show naive trial-level performance and then held-out-subject performance. Highlight the generalization gap.
1:10-1:35	Show the permutation evidence and top associated channel-band features. Explain that these are associations, not causal brain claims.
1:35-1:55	Open the risk report: sensitive information detected or inconclusive; subject dependence; unsupported claims; recommended remediation.
1:55-2:00	NerveML helps teams understand what neural data reveals and whether their models can be trusted before the data is shared or used to train AI.

17. Strategic Meaning of the Prototype

17.1 What success proves

- The founders can turn a neuroscience validity problem into a repeatable software workflow.
- Neural privacy can be demonstrated through measurable inference attacks rather than abstract fear.
- NerveML has a credible wedge that does not require mass-market BCI adoption.

- The team understands the distinction between model performance, scientific validity, and inferential privacy.
- The platform can expand by adding new sensitive attributes, grouping units, modalities, and remediation methods.

17.2 What success does not prove

- It does not prove a large market or willingness to pay.
- It does not prove that emotion can always be decoded from neural data.
- It does not establish regulatory compliance or clinical utility.
- It does not yet create a durable moat; the moat must emerge from standards, benchmarks, integrations, accumulated failure modes, and customer workflows.

18. Scientific, Ethical, Legal, and Product Risks

Risk	Why it matters	Mitigation in prototype
Emotion overclaiming	Could undermine scientific credibility and create ethical backlash	Use “self-reported valence label”; publish permitted and unsupported claims
Small or benchmark-specific effects	May not generalize to real customer data	Show uncertainty; seek multiple datasets and design partners
Confounding by EOG/EMG/movement	Could mislabel physiological artifacts as emotion	Add artifact-only and covariate baselines in the next version
Privacy theater	A single classifier cannot certify privacy	Describe scan as evidence under a specified attack model, not proof of safety
Compliance overclaiming	Legal obligations vary by jurisdiction and use	Do not market the MVP as certification; involve counsel for later frameworks
Consulting trap	Every dataset may require custom work	Standardize schema, tests, reports, and integration points
Too-early market	Few neural teams may have a budget today	Start with EEG/sleep/human-sensing workflows and validate actual procurement needs
Dual use	A detector could also help improve sensitive inference	Control access to advanced attacks and orient product toward validation/remediation

19. Rapid Build Plan

Stage	Deliverable
Day 1 - Core pipeline	Dataset loader; feature table; Random Forest/Logistic model; naive and GroupKFold metrics
Day 1 - Product layer	Risk rule function; audit_report.json; one result chart
Day 2 - Evidence	Permutation test; feature ranking; scientific language review

Day 2 - Interface	Streamlit input/config/results/report screens
Day 3 - Demo	Clean real run; two-minute recording; screenshots for YC
Day 3 - Customer loop	Send demo to 5-10 relevant researchers/startups and ask whether they would run it on their own data

20. Post-MVP Roadmap

Phase	Capabilities	Commercial learning
V0: Emotion demo	Valence probe; subject leakage; report	Can people immediately understand the problem?
V1: Dataset scanner	Identity, sleep, diagnosis, motor and artifact probes; richer schemas	Which inference risks are top-three pains?
V2: Design-partner pilots	Private deployment; custom grouping units; remediation experiments	Who owns budget and what triggers purchase?
V3: SDK and CI	CLI/API; automated regression scans; model/version comparison	Can validation become recurring infrastructure?
V4: Enterprise	Access control; audit history; governance workflows; private cloud	Can reports support procurement, clinical validation, and data exchange?
V5: Standard layer	Benchmarks, risk cards, certification partnerships	Can NerveML become the default neural-data validation standard?

21. Engineer Kickoff Checklist

10. Choose DEAP, SEED, or an already prepared multi-subject EEG feature dataset.
11. Confirm exact label definition and subject/trial metadata.
12. Implement one reproducible baseline before adding UI.
13. Use scikit-learn Pipeline to keep learned preprocessing inside folds.
14. Run both StratifiedKFold and GroupKFold using identical features and model family.
15. Persist fold-level metrics, predictions, and configuration.
16. Implement within-subject label permutation.
17. Return a stable JSON schema to the UI.
18. Build Streamlit around real outputs, not mocked scores.
19. Add clear scientific disclaimers and bounded claim language.
20. Record the demo only after one clean end-to-end run.

22. Suggested Output Schema

```
{
  "dataset": {
    "name": "DEAP",
    "modality": "EEG",
    "n_subjects": 32,
    "n_trials": 1280,
    "target": "binary_self_reported_valence"
  },
  "evaluation": {
```

```

    "trial_random_balanced_accuracy": 0.00,
    "trial_random_roc_auc": 0.00,
    "subject_grouped_balanced_accuracy": 0.00,
    "subject_grouped_roc_auc": 0.00,
    "generalization_gap": 0.00
  },
  "permutation_test": {
    "n_permutations": 100,
    "p_value": 0.00,
    "null_mean": 0.00
  },
  "risk_flags": {
    "possible_subject_dependence": false,
    "sensitive_inference_evidence": "inconclusive",
    "cross_subject_generalization": "weak"
  },
  "top_features": [],
  "permitted_claim": "",
  "unsupported_claims": [],
  "recommendations": []
}

```

23. Final Product Brief

Build this first

A simple application that loads a multi-subject EEG dataset, probes whether self-reported valence can be inferred, compares naive trial-level validation with held-out-subject validation, runs a permutation null, and produces a bounded neural-data risk report.

The prototype is valuable because it connects three concepts that are usually separated: neural decoding, scientific validity, and inferential privacy. The model itself is not the product. The product is the standardized process that tells a team what its data reveals, whether the result is trustworthy, what claims are supported, and what must be fixed before sharing or deployment.